

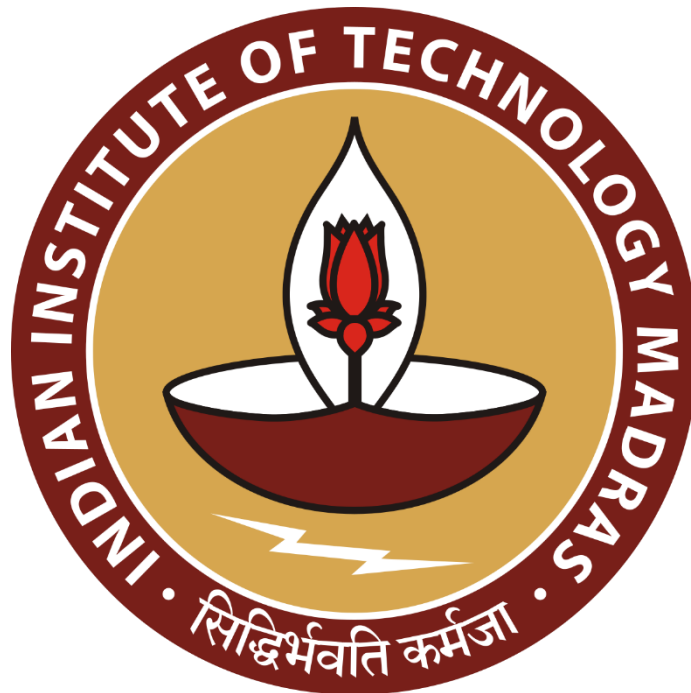
Financial Performance and Membership Retention Analysis of a Small-Scale Gym Using Data-Driven Methods

An End-Term report for the BDM capstone Project

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1 Executive Summary

This project focuses on analysing the financial and membership performance of a small-scale gym using data-driven methods. The gym primarily operates on membership-based revenue, offering plans such as monthly, quarterly, and personal training services. However, due to manual record-keeping practices, the business faced challenges in tracking payments, monitoring membership validity, and maintaining stable revenue.

The dataset used for this analysis was collected from the gym's physical registers and digitized into a structured Excel format. The data underwent cleaning and preprocessing to remove inconsistencies such as duplicate entries, missing member IDs, and irregular date formats. A structured payment validity logic was implemented to calculate membership expiry, and an analysis date was used to classify members into Active, At Risk, and Churn Risk categories. Additionally, Recency, Frequency, and Monetary (RFM) analysis was applied to segment members based on their engagement and contribution levels.

The analysis revealed significant fluctuations in monthly revenue, driven primarily by inconsistent membership renewals. A large proportion of members (63%) were identified as churn-risk, while only 21% remained active. RFM segmentation further highlighted that the majority of members fall into low-engagement categories, with a smaller proportion classified as high-value customers. Another key finding is that a significant share of revenue is generated from churn-risk members, indicating dependency on unstable customer segments.

Based on these findings, the project recommends implementing a structured digital tracking system to monitor membership validity and payment cycles. Improving customer retention through targeted engagement strategies, renewal reminders, and promotional offers is critical for stabilizing revenue. By adopting a data-driven approach to membership management, the gym can reduce churn, improve profitability, and achieve long-term business sustainability.

2 Detailed Explanation of Analysis Process & Method

This section explains the analytical framework used to transform manually maintained gym records into structured insights that support business decision-making. The analysis combines **quantitative spreadsheet-based methods** with **qualitative understanding obtained from discussions with the gym owner**.

The raw records were digitized and organized in **Microsoft Excel**, which served as the primary analytical tool for data preparation, statistical analysis, and visualization. The methodology focuses on converting operational transaction data into measurable indicators of revenue performance, membership activity, and payment continuity.

2.1. Data Preparation and Structuring

The dataset for this study was collected from the gym's physical registers, which record member payments and membership validity periods. Since the information was originally maintained in an unstructured manual format, the first step involved digitizing the records and organizing them into a structured spreadsheet.

During data preparation, the following steps were performed:

- Manual register entries were digitized into Microsoft Excel
- Date formats were standardized to enable time-based analysis
- Duplicate or incomplete entries were reviewed and corrected
- Payment records were validated against membership validity periods

2.2. Tools Used for Analysis

All analytical tasks were performed using **Microsoft Excel**, which provides sufficient computational and visualization capabilities for analysing operational data from small businesses.

The following Excel features were used:

- Statistical formulas such as **SUM, SUMIFS, COUNTIFS, AVERAGE, MAX, and MIN**
- **Pivot tables** were used as the primary tool for aggregation and dashboard visualization.
- **Line, column, and pie charts** for visualizing trends and comparisons
- **Date functions** for grouping transactions into monthly time periods

Excel was selected because it is widely accessible, easy to use, and suitable for the scale of analysis required for a small gym business.

2.3. Method for Monthly Revenue and Financial Analysis

To evaluate the gym's financial performance, a **time-series analysis approach** was used to analyse revenue trends over time.

Methodology:

- Payment dates were grouped by month using PivotTables.
- Monthly revenue was calculated by summing all payments received in each month using **SUMIFS** function.
- Fixed monthly expenses such as rent, electricity, and trainer fees were recorded separately.
- Monthly profit was calculated as:

$$\text{Profit} = \text{Total Monthly Revenue} - \text{Total Monthly Expenses}$$

Purpose:

This method enabled the identification of revenue trends, peak earning months, and periods of low income. By aggregating daily transactions into monthly summaries, the gym owner gains a clearer understanding of financial stability and business growth patterns.

2.4. Method for Membership Activity and Retention Analysis

Membership continuity was analysed by examining the number of active members during each month.

Methodology:

- Members whose membership validity covered a particular month were classified as **active members**.
- Monthly counts of active members were calculated using **COUNTIFS**.
- Membership trends were visualized to observe fluctuations in active member counts.

To evaluate membership stability, retention and churn indicators were calculated.

$$\text{Retention Rate} = (\text{Active Members} / \text{Total Members}) \times 100$$

$$\text{Churn Rate} = (\text{Members with Expired Memberships} / \text{Total Members}) \times 100$$

Purpose:

These indicators help assess member continuity and identify months with declining participation. A high churn rate indicates potential revenue instability and highlights the need for improved renewal tracking mechanisms.

2.5. Method for Payment Tracking and Membership Validity Analysis

One of the key operational challenges faced by the gym is tracking expired memberships and delayed payments.

Methodology:

- Membership expiry dates were compared with payment records
- Members were classified into two categories: **Active** and **Expired**.
- Logical date comparisons in Excel were used to identify overdue or unpaid memberships.

Purpose:

This analysis highlights weaknesses in manual payment monitoring and demonstrates how spreadsheet-based tracking can help identify revenue leakage caused by delayed membership renewals.

2.6. Service-wise Revenue Contribution Analysis

To understand the financial contribution of different services, payments were categorized based on membership type.

Methodology:

- Revenue from **Normal gym memberships** and **Personal Training (PT)** services was separated.
- PivotTables were used to calculate service-wise revenue totals.
- Comparative charts were generated to visualize the contribution of each service category.

Purpose:

This analysis helps identify high-value services and provides insights that can support pricing strategies, promotional activities, and service expansion decisions.

2.7. Qualitative Inputs from Business Owner

In addition to numerical analysis, informal discussions were conducted with the gym owner to understand operational challenges and contextual factors influencing the business.

The discussions focused on:

- Difficulties in tracking membership renewals manually.
- Challenges in identifying expired memberships.
- Seasonal variations in new member registrations.

These qualitative insights helped interpret the analytical results and ensured that the findings accurately reflect real-world operational conditions.

2.8. Justification of the Analytical Approach

The chosen methodology is appropriate for the following reasons:

- Gym revenue and membership activity vary over time, making **time-series analysis suitable**.
- Excel provides adequate analytical capabilities for small-scale business data.
- The dataset is relatively small and does not require advanced programming tools.

- Combining quantitative analysis with qualitative insights ensures practical business relevance

Overall, the methodology provides a structured framework for converting raw operational data into actionable insights that support business decision-making.

2.9. Data Integrity and Limitations

During the data collection process, it was observed that the gym does not maintain a **daily attendance register**. As a result, direct analysis of visit frequency and member attendance behaviour could not be performed.

To address this limitation, **renewal frequency** (the number of months a member maintained an active membership) was used as a proxy indicator of engagement. This approach ensures that behavioural insights remain grounded in verified financial transactions while acknowledging the operational constraints of manual record-keeping.

2.10. Method for Membership Lifecycle Classification

Membership status was determined using a data-driven approach based on membership validity rather than relying on inconsistent manual payment labels.

Methodology:

- Membership expiry date was calculated using:

$$\text{Expiry_Date} = \text{Month_Start_Date} + \text{Membership_Duration}$$
- Membership duration was standardized based on plan type:
 - Monthly = 30 days
 - 3 Months = 90 days
 - 6 Months = 180 days
- An analysis reference date (31-08-2025) was used.
- Members were classified into:
 - Active → $\text{Expiry_Date} \geq \text{Analysis Date}$
 - At Risk → Expiry within 30–60 days
 - Churn Risk → Expired membership

Purpose:

This method ensures consistent classification of members based on actual membership validity, overcoming errors present in manual record-keeping.

2.11. Method for RFM Analysis

RFM (Recency, Frequency, Monetary) analysis was used to segment members based on behavioural patterns.

Methodology:

- **Recency:** Number of days since last payment
Calculated as:
 $\text{Recency} = \text{Analysis Date} - \text{Last Payment Date}$
- **Frequency:** Total number of payments made by each member
- **Monetary:** Total amount contributed by each member
- Scores:
Each metric was converted into scores (1–5) using ranking:
 - Lower Recency → higher score
 - Higher Frequency → higher score
 - Higher Monetary → higher score
- Final RFM Score = R_Score + F_Score + M_Score
- Segmentation:
 - Champion → High R, F, M
 - Loyal → Moderate R, F, M
 - At Risk → Low Recency
 - Churn Risk → Low in all metrics

Purpose:

This method enables identification of high-value customers and helps design targeted retention strategies.

3 Results and Findings

This section presents the key results obtained from the analysis of the gym's membership, revenue, expense, and payment data. The findings are structured based on key business problems identified in the study, including revenue instability, membership retention, and payment tracking inefficiencies. Each subsection interprets the results in the context of these problems and is supported by descriptive statistics and visualizations generated using Microsoft Excel.

Problem 1: Revenue Instability

3.1. Monthly Revenue Trend Analysis

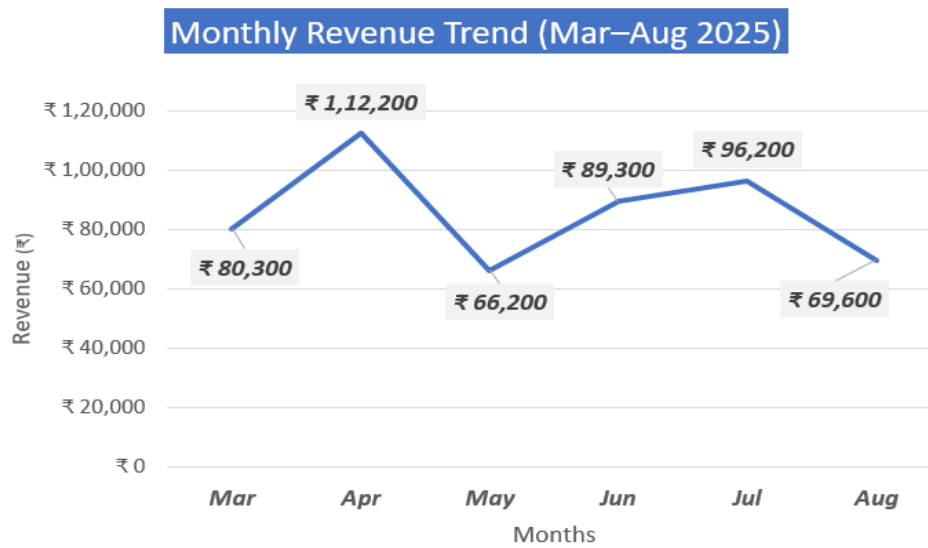


Fig. 1. Monthly Revenue Trend (₹)

The analysis of 378 payment transactions between March and August 2025 provides the first structured overview of the gym's financial performance. Digitizing manual payment records enabled the identification of monthly revenue patterns that were previously difficult to observe through physical registers.

Key Observations and Statistical Findings:

- **Total Revenue: ₹5,14,800**
 - The gym generated a total revenue of **₹5,14,800** during the observed six-month period.
 - This figure represents the first objective quantification of the business's gross income, providing a baseline for assessing long-term sustainability.
- **Average Monthly Revenue: ₹85,800**
 - On average, the gym earned approximately **₹85,800** per month.
 - This value reflects the typical monthly income level and serves as a vital benchmark for evaluating future growth or decline in membership engagement.
- **Maximum Monthly Revenue: ₹1,12,200 (April)**
 - The highest revenue was recorded in April, reaching **₹1,12,200**.
 - This peak indicates significantly stronger membership activity and improved renewal compliance during that specific month.
 - Statistically, this surge may be attributed to seasonal fitness motivation (post holiday or pre-summer goals) and more effective, manual payment follow-ups conducted by the owner during that window.

- **Minimum Monthly Revenue: ₹66,200 (May)**
 - The lowest revenue occurred in May, dropping to **₹66,200**.
 - This represents a sharp **41% decline** from the April peak, highlighting a period of mass renewal delays or member drop-offs.
 - Crucially, because operational fixed expenses remained unchanged at **₹62,000**, this revenue trough had a direct and severe negative impact on net profitability, leaving a margin of only ₹4,200.

Analysis of Financial Volatility:

Overall, the **wide variation** between the maximum (₹1,12,200) and minimum (₹66,200) revenue values exposes the gym's highly **unstable income pattern**. The business is currently vulnerable to "Revenue Sensitivity," where even a slight dip in renewals can expose the ability to cover fixed costs like rent and salaries.

This finding directly supports **Problem Statement 3.1**, emphasizing the urgent need for a systematic, digital revenue tracking system to enable accurate budgeting, financial forecasting, and proactive business planning.

Problem 2: Membership Retention and Activity

3.2. Membership Activity Analysis

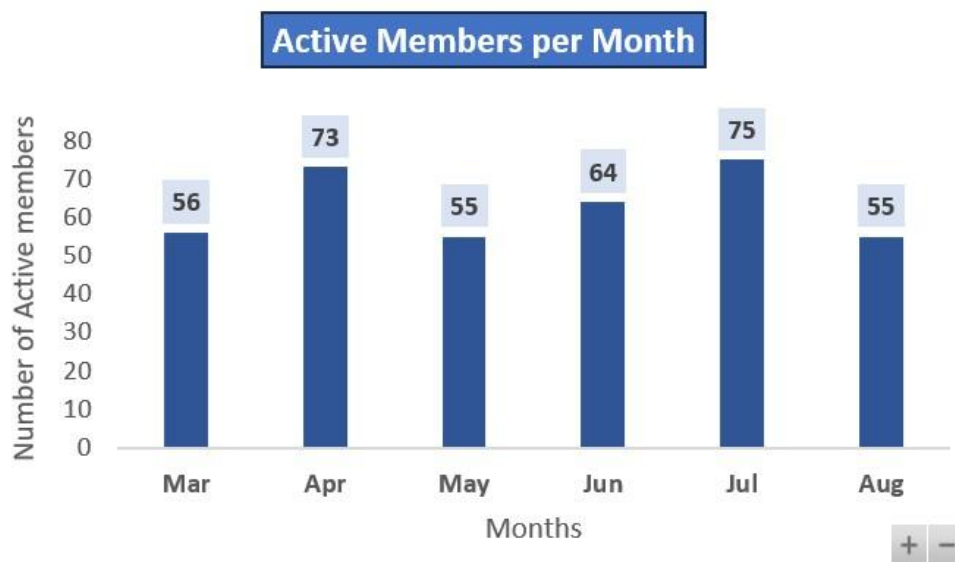


Fig. 2. Active Members per Month (Number of Members)

Membership activity analysis is fundamental to understanding the operational volume that drives the gym's financial performance. This section directly addresses the gym's historical inability to track member continuity, renewals, and long-term retention due to manual recordkeeping constraints.

Membership activity directly influences revenue stability because membership fees represent the primary source of income for the gym.

Key Statistical Findings from Membership Data:

- **Total Active Member Instances: 378**
 - Over the six-month study period, a cumulative total of **378 active member instances** was recorded.
 - This metric provides a high-level view of the gym's utilization rate across the analysed timeframe.
- **Average Monthly Active Members: 63**
 - On average, the gym maintained a base of **63 active members per month**.
 - While this indicates a consistent core community, the lack of an upward trend suggests that the business is in a state of stagnant growth, likely due to high churn offsetting new registrations.
- **Fluctuating Monthly Membership Levels**
 - The data reveals significant variation in active membership counts, ranging from a peak of **75 members** in July to a low of **55 members** in May and August.
 - These fluctuations are symptomatic of inconsistent renewal behaviour among the member base and highlight the total absence of structured, digital follow-up mechanisms to prompt expiring members.
- **Critical Lack of Retention Visibility**
 - Under the previous manual system, the gym owner could not distinguish between "Loyal" recurring members and "One-Time" short-term users.
 - Without this visibility, it was impossible to identify churn patterns (where a significant proportion of members are identified as churn risk (63%) based on lifecycle analysis in Section 3.5.) or implement rewards for long-term loyalty.

Managerial Implications:

The direct correlation between the number of active members and monthly revenue confirms that membership volume is the primary driver of the gym's financial viability. As shown in the "Danger Zone" months of May and August, a drop to **55 members** brings the business dangerously close to its break-even point.

This analysis directly addresses **Problem Statement 3.2**, demonstrating an urgent need for the proposed structured membership tracking system. Implementing such a system will allow the management to improve retention through data-backed insights, identify loyal customer segments, and ensure the long-term sustainability of the gym.

3.3. Expense Structure Analysis

Month	Rent	Electricity	Trainer Salary	Total Cost	Total Revenue	Profit (Net Income)	Profit Margin
March	₹ 30,000	₹ 20,000	₹ 12,000	₹ 62,000	₹ 80,300	₹ 18,300	23%
April	₹ 30,000	₹ 20,000	₹ 12,000	₹ 62,000	₹ 1,12,200	₹ 50,200	45%
May	₹ 30,000	₹ 20,000	₹ 12,000	₹ 62,000	₹ 66,200	₹ 4,200	6%
June	₹ 30,000	₹ 20,000	₹ 12,000	₹ 62,000	₹ 89,300	₹ 27,300	31%
July	₹ 30,000	₹ 20,000	₹ 12,000	₹ 62,000	₹ 96,200	₹ 34,200	36%
August	₹ 30,000	₹ 20,000	₹ 12,000	₹ 62,000	₹ 69,600	₹ 7,600	11%
Total	₹ 1,80,000	₹ 1,20,000	₹ 72,000	₹ 3,72,000	₹ 5,13,800	₹ 1,41,800	28%

Table 3.1 summarizes the monthly expense components, revenue and net profit values of the gym.

Understanding the cost side of the business is essential for evaluating the gym's financial resilience. This section breaks down the monthly fixed costs of Muscle and Fitness 2 and examines how they interact with fluctuating revenue streams to determine net profitability.

Core Insights from Expense Analysis:

- **Fixed Monthly Expenses: ₹62,000**
 - The gym operates with a fixed monthly expense of **₹62,000**, comprising rent (**₹30,000**), electricity (**₹20,000**), and trainer salaries (**₹12,000**).
 - This amount represents the minimum operational threshold that must be covered each month to avoid financial loss.
- **Rigid Consistency of Cost Structure**
 - The expense pattern remains constant across all months analysed, confirming a fixed-cost model with negligible variable costs.
 - While this ensures predictability in budgeting, it also limits the gym's ability to reduce expenses during periods of low revenue.
- **High Operating Leverage and Profit Impact**
 - Due to the fixed nature of costs, any decline in revenue has a disproportionately large impact on net profit.
 - For example, profit dropped from **₹50,200** in April to **₹4,200** in May, indicating a sharp decline of over **90%** in earnings despite a relatively moderate fall in revenue.
- **Profit Margin Volatility**
 - Profit margins fluctuate significantly across months, ranging from a high of **45%** in April to a low of **6%** in May.

- This variation reflects unstable financial performance and highlights the gym's dependence on consistent membership renewals.
- **Operational Risk in Low-Revenue Periods**
 - This fixed structure creates significant financial pressure during periods of poor renewal compliance.
 - In May, for example, after paying the ₹62,000 operational "hurdle," the business was left with only **₹4,200** in surplus, leaving no room for equipment maintenance or unexpected repairs.

Strategic Conclusion on Revenue Stability:

This analysis demonstrates that the gym operates under a high fixed-cost, high-risk financial model where revenue stability is critical for sustainability. Since expenses remain constant regardless of performance, maintaining consistent cash inflow becomes essential.

Therefore, implementing a digital payment tracking and reminder system is not an optional enhancement but a necessary strategy to ensure timely renewals and maintain revenue above the ₹62,000 operational threshold, thereby safeguarding profitability.

3.4. Profit Analysis

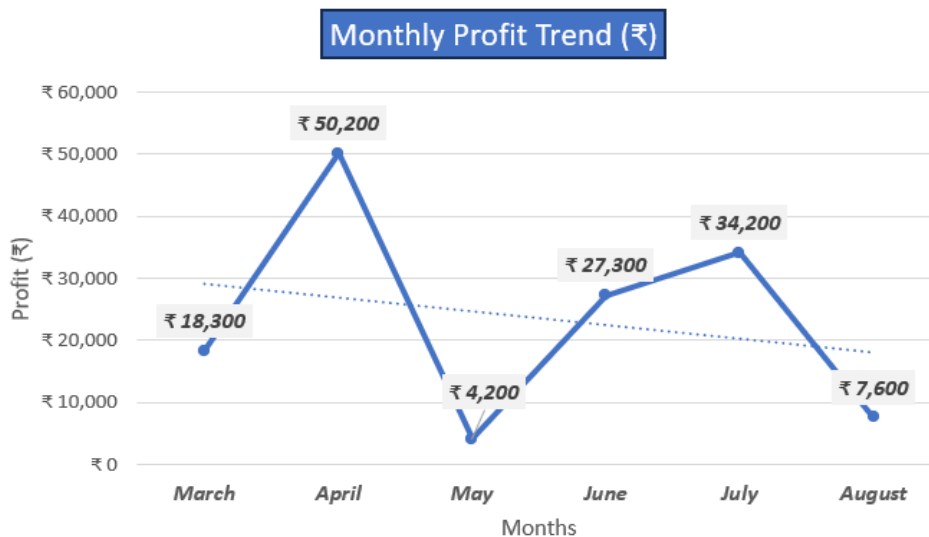


Fig. 3. Monthly Profit Trend (₹)

Profit analysis integrates revenue trends with the gym's fixed expense structure to evaluate overall financial sustainability. By subtracting monthly operational costs from revenue, the analysis provides a clear measure of the business's net profitability.

Core Financial Findings:

- **Highest Monthly Profit: ₹50,200 (April)**
 - The maximum profit was recorded in April, corresponding directly with the peak in membership renewals and revenue collection.
 - This peak confirms a strong direct relationship between revenue volume and profitability, proving that once the fixed "hurdle" of ₹62,000 is cleared, the business has high potential for surplus income.
- **Lowest Monthly Profit: ₹4,200 (May)**
 - The lowest profit occurred in May, leaving the gym with a razor-thin margin of survival.
 - This critical low was driven entirely by a reduction in membership renewals while the fixed operational expenses remained unchanged.
 - In this state, the business is highly vulnerable to "Financial Shock"; any unexpected equipment repair or utility spike would have resulted in an immediate net loss.
- **Revenue-Driven Profit Fluctuations**
 - The analysis proves that profit variations are influenced almost exclusively by changes in membership payment behaviour rather than fluctuations in operational costs.
 - Due to the high **Operating Leverage** of the business, a minor decline in active members (from 73 in April to 55 in May) resulted in a disproportionate **91% drop** in net profit.

Managerial Implications:

The profit trendline clearly demonstrates that the gym currently operates with an unstable safety margin. To ensure consistent profitability above the ₹30,000 threshold, the business must stabilize its monthly active membership base at **67 members or higher**, as further proved by the Correlation Analysis in Section 3.7. This visibility into profit behaviour directly supports improved financial decision-making and underscores the importance of the digital tracking system.

Problem 3: Payment Tracking and Churn Behaviour

3.5. Membership Retention and Churn Analysis

Distribution of Members by Membership Status

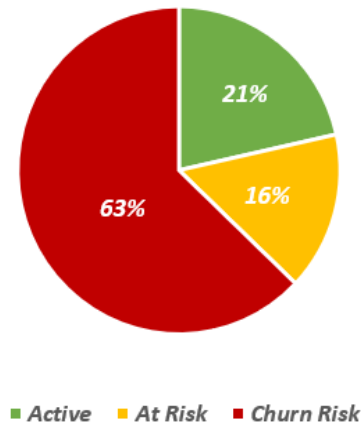


Fig. 4. Distribution of Members by Membership Status (%)

Membership retention analysis provides critical insight into the continuity of gym members and the stability of the gym's revenue stream. Member status in this study is determined using a structured, data-driven method based on membership expiry dates relative to the analysis date. Based on this lifecycle classification, members were grouped into three categories: Active, At Risk, and Churn Risk.

Core Analytical Findings

- **Active Members: 21% (81 members)**
 - Approximately 21% of members were classified as active during the analysis period.
 - These members have valid memberships beyond the analysis date, indicating ongoing engagement and consistent renewal behaviour.
- **At Risk Members: 16% (59 members)**
 - These members are close to their membership expiry date and have not yet renewed, indicating a high probability of churn if no action is taken.
- **Churn Risk Members: 63% (237 members)**
 - A significant proportion of members (63%) fall under the churn-risk category.
 - These members have already exceeded their membership validity period and have not renewed within the expected timeframe, indicating disengagement from the gym.
- **Implications for Revenue Stability**
 - Since gym revenue primarily depends on recurring membership payments, the high proportion of churn-risk members poses a serious threat to financial stability. With only 21% of members actively contributing to current revenue, the business is heavily exposed to fluctuations and periods of low income.

- The presence of a considerable at-risk segment (16%) further indicates that without timely intervention, the churn percentage may increase even further, worsening revenue instability.

Managerial Implications

The findings clearly highlight the need for a structured and proactive membership tracking system. The current high churn level indicates gaps in monitoring renewal cycles and lack of systematic follow-ups.

Implementing an Excel-based tracking system with expiry alerts and automated monitoring can significantly improve visibility into member status. Additionally, targeted retention strategies such as reminder notifications, limited-time renewal offers, and personalized engagement can help convert at-risk and churn-risk members into active members.

Even a small improvement in retention—for example, converting 10–15% of churn-risk members into active members—can lead to a substantial increase in monthly revenue without requiring additional customer acquisition.

Problem 4: Revenue Diversification Opportunity

3.6. Service-Wise Revenue Distribution Analysis (₹)

3.6.1. Revenue Contribution by Service Type

The gym's revenue model relies on two distinct service streams. Understanding the "Service Mix" is essential for identifying which programs drive the most value versus which programs provide the highest volume.

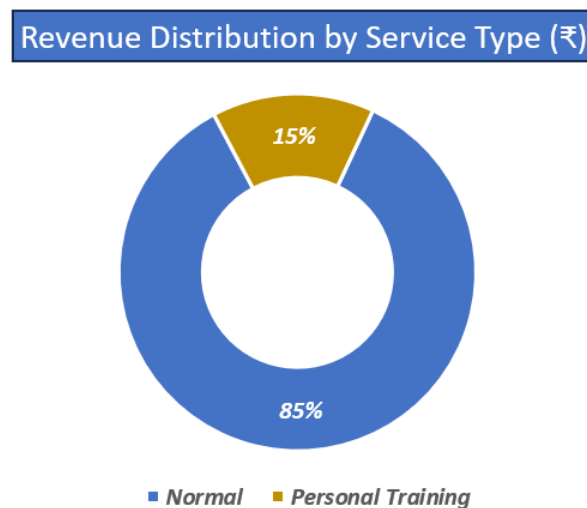


Fig. 5. Service-Wise Revenue Distribution (₹)

This analysis illustrates the financial weight of the two primary service streams offered by Muscle and Fitness 2 Gym. By visualizing the revenue mix, we can identify which services

provide the volume necessary for operational stability and which offer the highest potential for margin expansion.

Core Analytical Insights:

- **Normal Membership Revenue: ₹4,38,300**
 - The vast majority of the gym's total revenue (**approx. 85%**) is generated through standard memberships.
 - This indicates that the business relies heavily on its "Normal" plans as the primary engine for covering fixed costs like rent and electricity.
- **Personal Training Revenue: ₹75,500**
 - Personal Training contributes a significantly smaller share of the total revenue (**approx. 15%**).
 - While Personal Training is a premium, higher-priced service, the data reveals that it is currently under-utilized by the broader member base.
- **Revenue Concentration Risk**
 - The heavy dependence on standard memberships exposes the gym to a "single point of failure" risk.
 - If a seasonal dip or churn event affects normal memberships, the business lacks a strong secondary revenue stream to provide a financial buffer.
- **Strategic Growth Opportunity**
 - The relatively low 15% contribution from PT services represents a massive untapped growth opportunity.
 - Because Personal Training services provide higher revenue per member, increasing this segment's share through bundled offers or targeted promotions could significantly boost net profitability without requiring additional capital expenditure.

Managerial Conclusion:

This distribution analysis provides the gym owner with a data-driven map for business expansion. Rather than focusing solely on acquiring new "Normal" members, the owner can stabilize the business by upselling existing loyal members into Personal Training programs, effectively diversifying the revenue mix and increasing the average revenue per member (ARPM).

3.7. Member Segmentation Using RFM Analysis

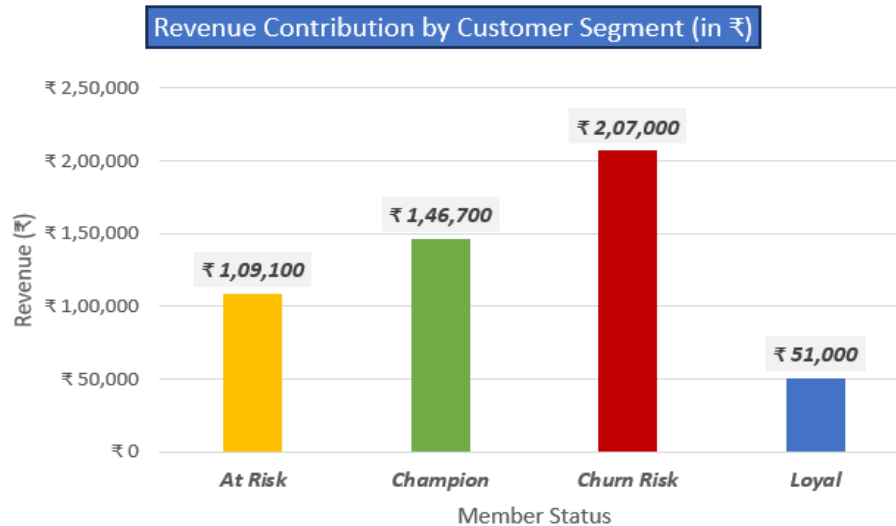


Fig. 6. Revenue Contribution by Customer Segment (₹)

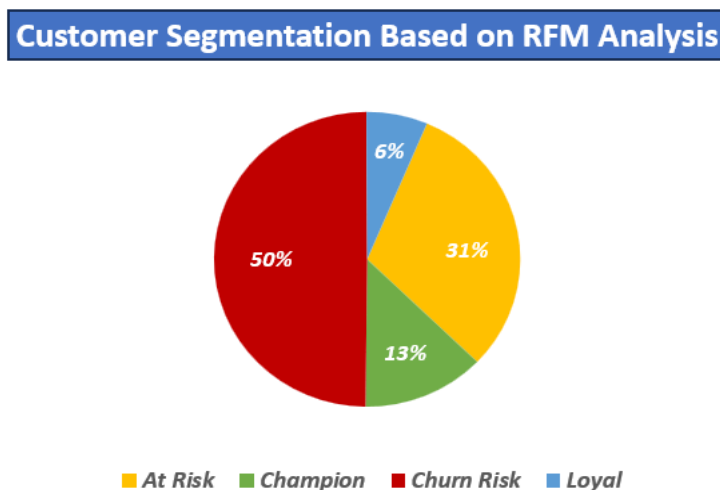


Fig. 7. Customer Segmentation Based on RFM Analysis (%).

To gain deeper insights into member behaviour and engagement patterns, **RFM (Recency, Frequency, Monetary)** analysis was conducted on the gym's payment dataset. This analytical approach evaluates members based on three key dimensions: how recently they renewed their membership (**Recency**), how frequently they made payments (**Frequency**), and the total monetary contribution generated by each member (**Monetary**).

Based on these metrics, members were segmented into four behavioural categories: **Champions**, **Loyal Members**, **At-Risk Members**, and **Churn-Risk Members**. This segmentation provides a clearer understanding of customer engagement levels and highlights the contribution of each segment to overall revenue.

Core Analytical Findings

- **Champion Members: 26 members (13%)**

- Champion members represent the most valuable segment of the gym's customer base. These members have high recency, frequency, and monetary scores, indicating strong engagement and consistent renewal behaviour.
- Revenue Contribution: ₹1,46,700

• **Loyal Members: 13 members (6%)**

- Loyal members demonstrate stable engagement with moderate frequency and spending patterns. While smaller in number, they contribute consistently to the gym's revenue.
- Revenue Contribution: ₹51,000

• **At-Risk Members: 62 members (31%)**

- At-risk members show declining engagement, often characterized by increasing gaps between renewals. Without timely intervention, these members are likely to transition into churn-risk customers.
- Revenue Contribution: ₹1,09,100

• **Churn-Risk Members: 100 members (50%)**

- Churn-risk members represent the largest segment in the dataset. These members have low recency and reduced engagement levels, indicating a high probability of discontinuing their membership.
- Revenue Contribution: ₹2,07,000

Key Insight: Revenue Concentration Risk

- A critical observation from the analysis is that **churn-risk members contribute the highest share of revenue (₹2,07,000)** despite being the least engaged segment. This indicates a significant dependency on unstable customers, which poses a major risk to long-term revenue sustainability.
- At the same time, high-value segments such as Champions and Loyal members contribute comparatively less, highlighting the need to strengthen long-term customer retention strategies.

Managerial Implications

The RFM segmentation highlights the importance of adopting differentiated customer management strategies:

• **Champion and Loyal Members**

These segments should be retained through loyalty programs, personalized engagement, and premium services such as personal training. Strengthening relationships with these members can improve long-term revenue stability.

• **At-Risk Members**

These members require proactive engagement strategies such as renewal reminders, limited-time offers, and targeted communication to prevent them from moving into churn-risk.

- **Churn-Risk Members**

This segment requires reactivation strategies such as discounts, rejoining incentives, and personalized outreach. Reducing churn in this segment can significantly improve revenue consistency.

Strategic Conclusion

The RFM-based segmentation reveals a critical imbalance in the gym's customer base, where a large portion of revenue depends on members with low engagement. By shifting focus towards retaining high-value customers and converting at-risk members into active users, the gym can move towards a more stable and sustainable revenue model

3.8. Overall Business Insights

The analysis of membership data, revenue trends, cost structure, and customer segmentation provides a clear understanding of the gym's operational and financial performance. The findings directly address key business challenges related to revenue instability, member retention, and inefficient tracking systems.

- **Revenue instability:**

- Monthly revenue shows significant fluctuations, with higher earnings in months like April and July and lower performance in May and August. This variation is primarily driven by inconsistent membership renewals, indicating the absence of a structured monitoring system.

- **Cost Structure and Profit Sensitivity:**

- The gym operates with fixed monthly expenses of ₹62,000. Since costs remain constant, profit levels vary directly with revenue, ranging from ₹50,200 (April) to ₹4,200 (May). This highlights that profitability is highly sensitive to changes in membership activity.

- **High Churn and Retention Issue:**

- A large proportion of members (63%) fall under churn risk, while only 21% are active and 16% are at risk. This indicates weak retention mechanisms and a high likelihood of continued revenue instability if no corrective actions are taken.

- **RFM-Based Customer Imbalance:**

- RFM analysis reveals that 50% of members are classified as churn-risk, whereas only 13% are champions and 6% are loyal. A critical insight is that churn-risk members contribute the highest share of revenue, exposing the business to dependency on low-engagement customers.

- **Need for Structured Tracking System:**

- The initial dataset highlighted inconsistencies in payment tracking due to manual record-keeping. The absence of a systematic approach to monitor membership validity and renewals leads to delayed actions and potential revenue loss.

4 Interpretation of Results and Recommendations

4.1. Financial Overview of the Business

- The analysis shows that the gym operates with a fixed monthly cost of ₹62,000, including rent, electricity, and trainer salaries. Due to this fixed cost structure, profitability is highly dependent on consistent membership renewals and revenue generation.
- The revenue trend indicates noticeable fluctuations across months, which directly impact profit levels. Profit ranges from a high of ₹50,200 to a low of ₹4,200, highlighting the unstable nature of the gym's financial performance. Since expenses remain constant, even a small decline in revenue leads to a sharp reduction in profit margins.
- The profit trend further indicates that certain months (April and July) perform significantly better, while others (May and August) experience lower profitability. This inconsistency reflects dependence on irregular membership renewals.

Conclusion:

Maintaining a stable base of active members is essential to ensure consistent profitability. Based on the observed revenue patterns, sustaining approximately 65–70 active members is critical to avoid operating near the break-even point.

4.2. Membership Behaviour Interpretation

The updated membership lifecycle analysis shows that:

- **21% of members are Active**
- **16% are At Risk**
- **63% fall under Churn Risk**
- This indicates a significantly high level of customer attrition, with only a small proportion of members consistently renewing their memberships.
- The presence of an “At Risk” segment further highlights that a portion of the customer base is close to disengagement and may soon transition into churn if no action is taken.
- This pattern suggests the absence of a structured system to monitor membership validity and renewal behaviour. As a result, the gym faces challenges in maintaining a stable and predictable revenue stream.

Conclusion:

Improving member retention is more critical than acquiring new customers. Converting even a small proportion of at-risk and churn-risk members into active members can significantly improve revenue stability.

4.3. Customer Segmentation Interpretation (RFM Analysis)

The RFM analysis provides deeper insights into customer engagement patterns:

- **50% of members are classified as Churn Risk**
- **31% are At Risk**
- **13% are Champions**
- **6% are Loyal Members**
- A key observation is that **churn-risk members contribute the highest share of revenue (₹2,07,000)**. This indicates that a large portion of the gym's revenue depends on customers with low engagement and high probability of discontinuation.
- In contrast, high-value segments such as Champions and Loyal members contribute relatively less to total revenue, highlighting an imbalance in the customer base.
- This situation creates a **revenue concentration risk**, where the business depends heavily on unstable customers rather than a strong base of loyal members.

Conclusion:

The gym must shift its focus from short-term revenue generation to long-term customer retention by strengthening relationships with high-value members and improving engagement among at-risk segments.

4.4. Payment Monitoring Interpretation

- The analysis highlights significant gaps in payment tracking due to reliance on manual record-keeping. The presence of a large number of expired memberships indicates weak monitoring of renewal cycles and delayed follow-ups.
- Without a structured system, it becomes difficult to track due dates, identify inactive members, and ensure timely renewals. This leads to inefficiencies in payment collection and increases the risk of revenue leakage.
- The findings clearly show that manual tracking is not sufficient for managing a growing member base and results in operational challenges.

Conclusion:

A structured, data-driven payment tracking system is necessary to monitor membership expiry, improve renewal rates, and ensure timely collection of payments.

4.5. Recommendations

- **Short-Term (Immediate Actions)**
 - Implement the Excel-based tracking system developed in this project.
 - Monitor membership expiry dates using automated alerts and conditional formatting.
 - Maintain structured digital records of all transactions.
 - Identify and contact at-risk members before expiry.
- **Medium-Term (Operational Improvements)**

- Regularly track key metrics such as revenue, active members, and profit.
 - Develop targeted engagement strategies for at-risk members.
 - Strengthen retention of Champion and Loyal members through personalized services
 - Monitor break-even membership levels to avoid low-profit periods
- **Long-Term (Strategic Improvements)**
 - Introduce loyalty programs and incentives to improve long-term retention.
 - Develop re-engagement campaigns for churn-risk members.
 - Adopt a fully data-driven approach to membership and financial management.
 - Expand value-added services to increase revenue per member.

4.6. Expected Business Impact

The implementation of the recommended data-driven strategies is expected to significantly improve the gym's operational efficiency and financial performance.

- **Improved Revenue Stability:**

- Tracking renewals and managing member engagement will reduce fluctuations in monthly revenue and ensure consistent income.

- **Increased Member Retention:**

- Targeted strategies for at-risk and churn-risk members will improve renewal rates and reduce customer attrition.

- **Reduction in Revenue Risk:**

- By reducing dependency on churn-risk customers, the business can achieve a more stable and sustainable revenue structure.

- **Higher Revenue per Member:**

- Better retention and consistent revenue will lead to improved profit margins despite fixed costs.

- **Better Decision-Making:**

- Continuous monitoring of key metrics will enable informed and timely business decisions.

5 Additional Information

The complete project files, including the cleaned dataset, RFM analysis, pivot tables, and dashboard charts, have been shared through the Google Drive link below:

Link: [Business Data Management \(BDM\) Project Folder](#)

Note: The analysis has been performed in Microsoft Excel. For accurate visualization of pivot tables and charts, it is recommended to open or download the file using Microsoft Excel. Google Sheets preview may not fully support all Excel features.